Numerical Stability of Explicit and Implicit Methods for a Bounded Memristor ODE

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1 Overview

This project studies the numerical behavior of three single-step methods applied to a nonlinear memristor initial value problem. The objective is not only to compute a trajectory, but to compare how each method reacts when the time step becomes coarse relative to the time scale of the excitation. The main quantities of interest are the internal state trajectory, violations of the physical state bounds, and the resulting voltage-current hysteresis loops.

2 Memristor Model

The device is modeled using the linear ion drift HP TiO2 memristor model. The internal state variable $w(t)$ represents the width of the doped region. It is physically constrained by

$$0 \leq w(t) \leq D,$$

where D is the device thickness.

The memristance is

$$M(w) = R_{\text{on}}\frac{w}{D} + R_{\text{off}}\left(1 - \frac{w}{D}\right),$$

with applied voltage

$$v(t) = V_0 \sin(\omega t).$$

The current is then

$$i(t, w) = \frac{v(t)}{M(w)}.$$

The state equation solved numerically is

$$\frac{dw}{dt} = \frac{\mu_v R_{\text{on}} v(t)}{D^2 M(w)}.$$

3 Numerical Methods

The initial value problem was solved using Explicit Euler, fourth-order Runge-Kutta, and Implicit Euler. Explicit Euler advances the state using the derivative at the current time. RK4 uses four slope evaluations per step to obtain a higher-order explicit approximation. Implicit Euler advances the state using the derivative at the next time, which requires solving a nonlinear scalar equation at every step.

For Implicit Euler, the update satisfies

$$w_{n+1} = w_n + hf(t_{n+1}, w_{n+1}),$$

or equivalently

$$F(w_{n+1}) = w_{n+1} - w_n - hf(t_{n+1}, w_{n+1}) = 0.$$

This scalar nonlinear equation is solved by Newton iteration using an Explicit Euler prediction as the initial guess.

4 Experiment Design

The notebook evaluates all methods over the same time interval and with the same initial condition. Several step sizes are used, ranging from fine to deliberately coarse. Raw trajectories are preserved so that unphysical overshoot can be measured directly. A clipped copy of each trajectory is also computed for voltage-current plots, because the physical memristor interpretation requires w to remain in $[0, D]$.

The comparison uses four output types:

- state trajectory plots $w(t)/D$,
- overshoot summaries measuring how far raw states leave the interval $[0, D]$,
- V-I hysteresis loops computed from clipped states,
- runtime summaries measuring median cost per step over repeated solver calls.

5 Results

Figure 1 compares the raw state trajectories. Fine step sizes keep the methods close to one another. As the step size increases, the explicit methods become more sensitive to the discretization, and bound violations become easier to observe.

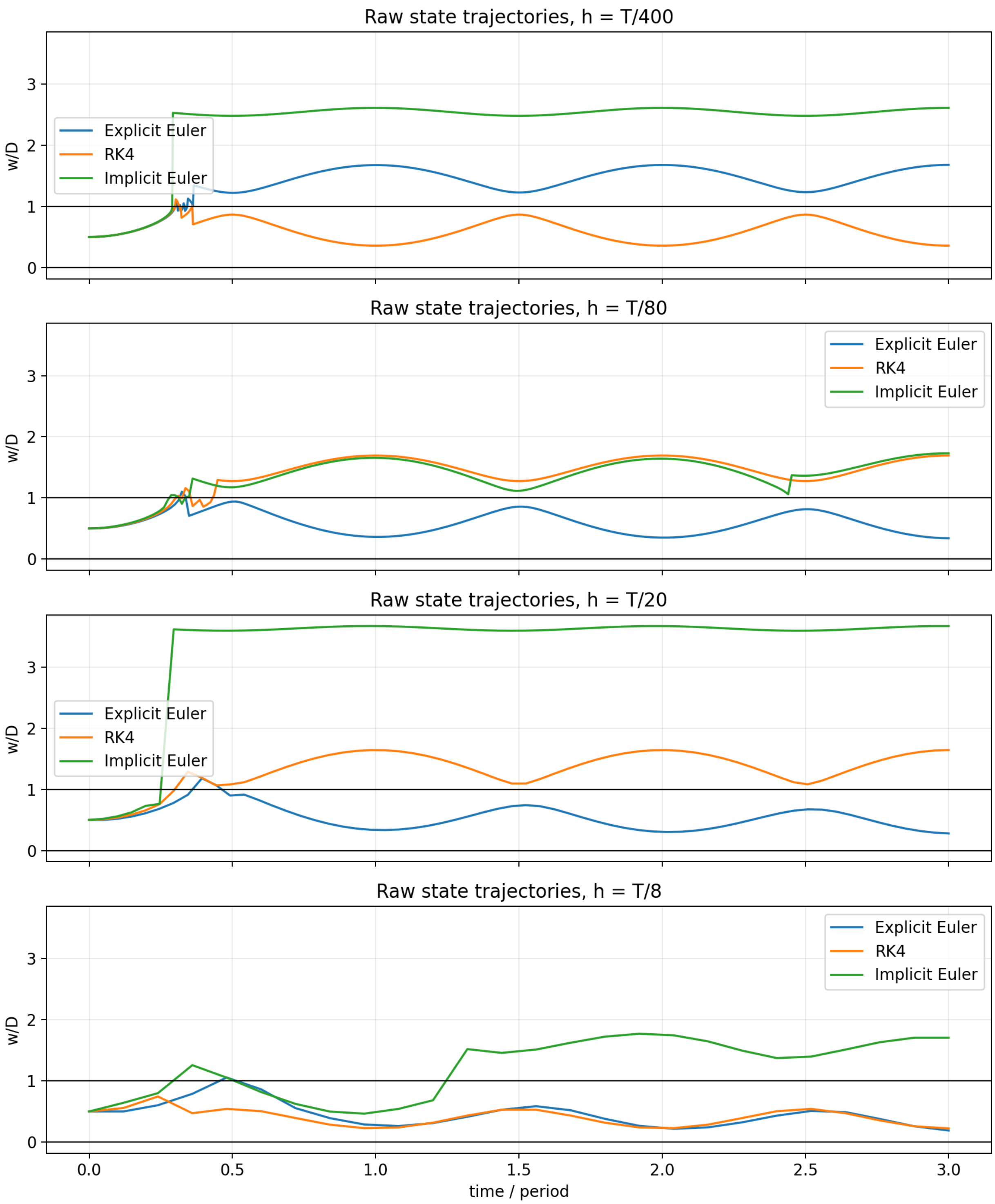

Figure 1: Raw normalized state trajectories for the tested methods and step sizes.

Figure 2 summarizes boundary violations. This view makes the physical constraint explicit and helps distinguish a numerically smooth trajectory from one that is physically meaningful.

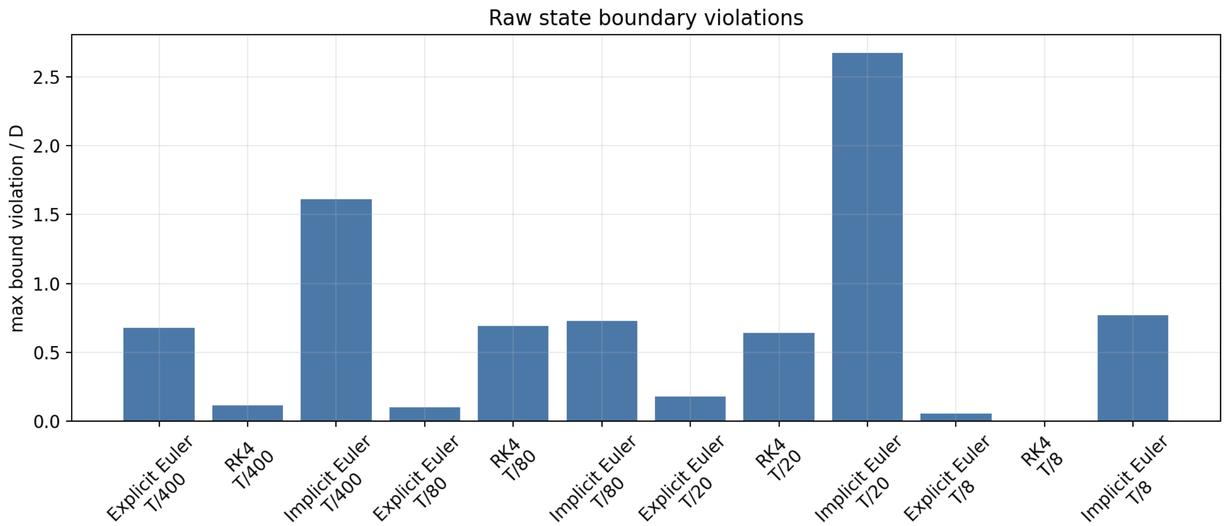

Figure 2: Raw state bounds compared against the physical interval $0 \leq w \leq D$.

Figure 3 shows the resulting voltage-current curves using clipped physical states. Clipping is used only for this physical visualization; the overshoot tables are computed from raw trajectories.

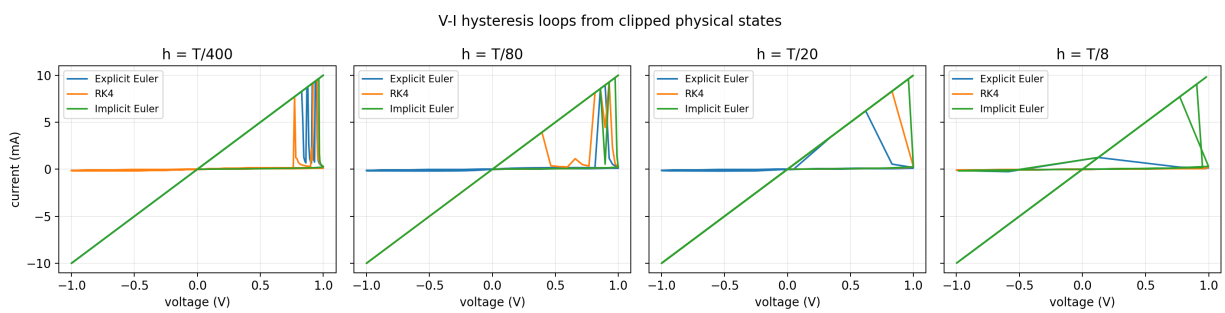

Figure 3: Voltage-current hysteresis loops computed from clipped physical states.

Figure 4 reports relative runtime per step. These timings are not hardware-independent benchmarks, but they show the expected computational-cost pattern: Explicit Euler is cheapest per step, RK4 pays for four slope evaluations, and Implicit Euler pays for nonlinear solves.

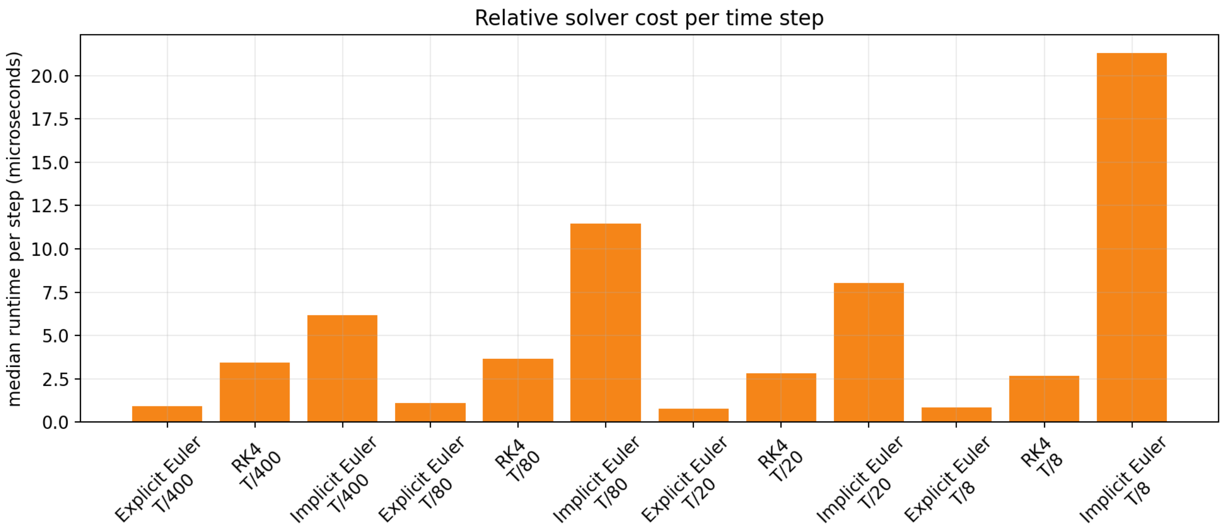

Figure 4: Median runtime per step over repeated solver calls.

6 Discussion

The project illustrates the tradeoff between accuracy, stability, and computational cost. Explicit Euler is simple but strongly step-size dependent. RK4 is more accurate for smooth trajectories at a given step size, but it is still explicit and can exhibit unphysical behavior when the step size is too coarse. Implicit Euler is only first-order accurate, and its future-time derivative can improve damping for stable modes, but it does not automatically enforce the memristor bounds. In this bounded nonlinear model, raw implicit updates can still leave the physical interval unless bound handling is added separately.

For this memristor problem, physical interpretation matters as much as numerical smoothness. A method can produce a curve that appears computationally well behaved while still violating $0 \leq w \leq D$. Preserving raw trajectories and separately clipping them for physical current calculations makes this distinction visible.

7 Conclusion

Explicit Euler, RK4, and Implicit Euler provide different views of the bounded memristor IVP. RK4 is generally the most accurate explicit method among the three when the step size is sufficiently small. Explicit Euler is the simplest and is visibly step-size dependent. Implicit Euler requires non-linear solves and may improve stability for some dynamics, but it is not a substitute for physical state constraints. The results support the proposal's central point: step-size choice, method behavior, and boundary treatment strongly affect whether the simulated memristor remains physically meaningful.